

Autoresearch Loops: A Causal Study of the Instruction File

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Abstract

An autoresearch loop is a system in which a language model proposes, executes and interprets experiments without a human in the loop. In the form popularized by Karpathy’s `autoresearch`, the whole experimental procedure is written in English in a single instruction file, `program.md`. This thesis treats the instruction file as a configuration vector. Five components of `program.md` are each written at two levels, all $2^5 = 32$ combinations are generated and executed on two classification tasks with twenty replicates each, giving 1,280 runs, and four behavioral metrics read from the structured transcripts are regressed on an orthogonal ± 1 coding of the components with a dataset fixed effect. Because the configurations are constructed instead of observed, the coefficients admit a causal reading, and because the factorial design is complete, the independence of the components can be tested. The results show that the components determine the process of the search rather than its outcome. The reporting style has the largest effect in the design and reduces the wasted trial ratio from 0.106 to 0.031. The stopping rule is the only component with an effect on either outcome metric, and its effect on the gain over default is smaller than the difference between the two datasets.

1 Introduction

1.1 Background

In an autoresearch loop, the experimentation itself is performed by a language model. The model is given a goal, a body of code it may edit, and an executor that runs a trial and returns a scalar result. From that point it proceeds autonomously: it proposes a change, executes it, reads the outcome, decides what to attempt next, and repeats until the budget is exhausted. The current meaning of the term was fixed by Karpathy’s `autoresearch` [5], which appeared in March 2026 and has since been forked several thousand times.

The architecture is a contract between three files. `prepare.py` prepares the data and defines the score; it is modified by no one, so that every experiment is scored identically. `train.py` contains the model and the training loop and is edited only by the agent, while `program.md` contains the instructions and is edited only by the human. In Karpathy’s original setting each run gets five minutes, so trials are equally costly [2].

The experiment cycle exists only as English prose within `program.md`, and the model itself is the execution layer. `program.md` is therefore not configuration supplied to a controller; it *is* the controller.

1.2 Motivation and Research Objective

An instruction file settles several matters simultaneously: what counts as success, how widely the agent may search, when it must stop, how it must report, and where effort should be concentrated. Two differently worded files therefore differ in all five at once, and no individual effect can be attributed to any one of them. Separating them requires a file in which one matter can be changed while the

rest are held fixed, which is to treat it not as prose but as a configuration vector. Five components of `program.md` are accordingly each written at two levels, and all $2^5 = 32$ combinations are generated and executed. The aim is not to propose a superior autoresearch loop, but to establish what an existing one is responding to.

Every comparison is between two configurations of the same system; no human baseline is invoked. The following research questions are raised:

RQ1 Which components of `program.md` causally determine which aspects of the loop’s research *process*, and how large is each effect?

RQ2 Do the components act independently of one another, or does the effect of one depend on the level of another?

1.3 Structure of the Thesis

This thesis is organized into four sections. Following this introduction, Section 2 describes how the instruction file is turned into a configuration vector, and explains why executing every configuration identifies the effect of each component. Section 3 presents the experimental setup, the metrics and the regression model, followed by the results. Section 4 concludes by summarizing the findings. Appendix A reproduces the three files the loop runs on, including the complete specification from which every instruction file of the study can be reconstructed.

2 Background

Figure 1 shows the control flow of one run.

2.1 The Instruction File as a Configuration

Every instruction file is generated from a single template. The fixed part specifies the research question, the executor interface and the JSON response format. The variable part consists of five paragraphs, each written at one of two levels; these five slots constitute the independent variables. Nothing else in the file varies. A file is described completely by

$$c = (c_1, \dots, c_5), \quad c_i \in \{0, 1\},$$

where the slots are taken in the order metric, breadth, stopping rule, output format and emphasis, yielding $2^5 = 32$ files. Concretely, $c = (1, 0, 1, 1, 0)$ is the file `M1-B0-S1-O1-E0`: the level-1 paragraph in the evaluation slot, level 0 in breadth, level 1 in stopping, level 1 in reporting and level 0 in emphasis. Its neighbor `M1-B0-S1-O0-E0` is the same file with one paragraph changed and all else fixed. The remaining thirty files are the other combinations.

Table 1 lists the levels. A production `program.md` hardcodes the baseline to be exceeded, specifies how failures are to be handled, fixes a reporting discipline and imposes a parsimony rule [2]. The five components in the table are drawn from those categories. One of them matters more than it looks: the loop carries no state except its own transcript. What the model wrote in the hypothesis and interpretation fields of the previous turns is the only thing it can read before proposing the next experiment. The reporting style *O* therefore determines how much of the loop’s reasoning is written down and made available to itself.

The ten paragraphs were written by a language model other than the agent. They are reproduced in Appendix B.2.

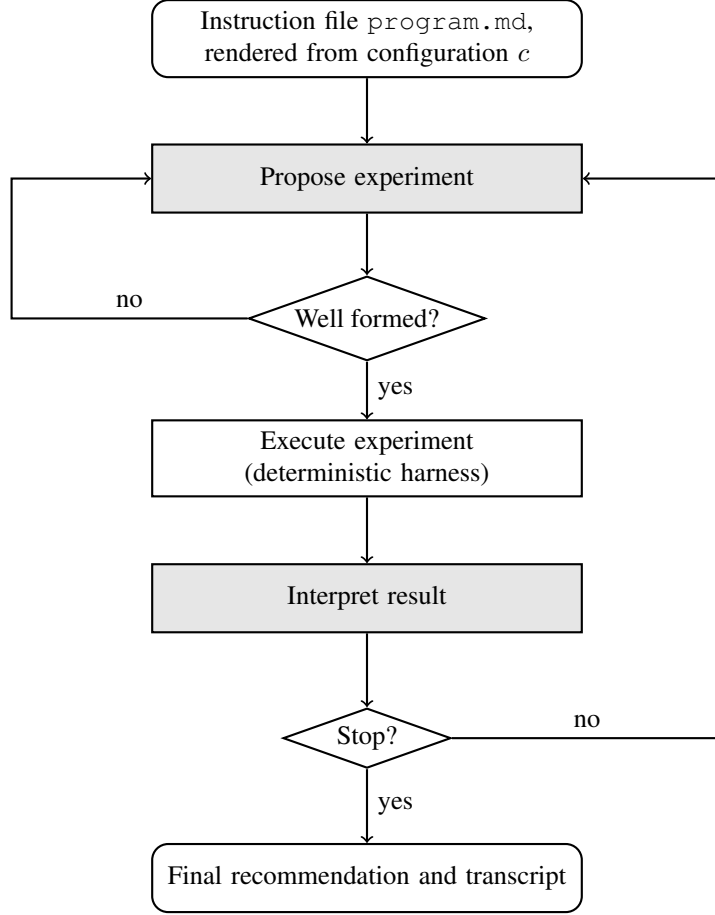


Figure 1: Control flow of one run. Shaded boxes are language-model calls. The unshaded box is deterministic, and hence every number the loop reasons about was measured and not generated. The validity check combines two checks in the code. The harness parses the reply and checks its shape, and `train.py` checks the whitelist before any estimator is fitted. A proposal that fails either check goes back to the proposal step. It is logged as an attempt, which is what the wasted trial ratio counts, and costs a turn but not an experiment. A run ends on the model’s stop decision, on eight executed experiments, or on four rejected or malformed replies, whichever comes first.

3 Experiment Setup and Results

All experiments were implemented in Python; the files the loop runs on are reproduced in the appendices.

3.1 Tasks, Executor and Baseline

Every configuration is executed against two canonical classification datasets, selected to differ in size, dimensionality and number of classes: Breast Cancer Wisconsin (569×30 , two classes) and Wine (178×13 , three classes). The loop may select either `RandomForestClassifier` or `GradientBoostingClassifier` under a fixed hyperparameter whitelist that the agent is not shown, given in Appendix D. Anything outside it is rejected before an estimator is built.

A single reference quantity is computed per dataset before any run, by a procedure involving no language model. The quantity $a_0(d)$ is the accuracy of a default-hyperparameter fit, taken as the better of the two model families, and every run is measured against it. On both datasets the better family is the random forest, with $a_0 = 0.9598$ on Breast Cancer and $a_0 = 0.9600$ on Wine on the cross-validated scale.

Table 1: The five varied components of `program.md`. A configuration is one choice of level per row. The last column gives the token count of each level’s paragraph, level 0 / level 1; the levels are not length-matched.

Code	Component	Level 0	Level 1	Tokens
<i>M</i>	Evaluation criterion	Success left undefined; the loop judges quality as it sees fit.	Success defined as five-fold cross-validated accuracy, with the across-fold standard deviation used only to break ties.	10 / 56
<i>B</i>	Search breadth	Commit to one model family for the session.	Compare both families within a session and vary several hyperparameters at once.	32 / 26
<i>S</i>	Stopping rule	Fixed budget of exactly three experiments.	Adaptive budget of up to eight, halted when an experiment fails to improve by more than 0.002.	32 / 59
<i>O</i>	Reporting style	Terse: the action taken and the numbers involved.	Full sentences stating hypothesis, expectation and interpretation.	20 / 34
<i>E</i>	Search emphasis	Explore first: several different configurations before refining any.	Exploit first: refine a promising configuration immediately.	24 / 32

3.2 Design and Budget

The design is the complete 2^5 factorial crossed with the task suite and replicated R times, giving $N = 2^5 \cdot D \cdot R$ runs. With $D = 2$ and $R = 20$, this yields $N = 1,280$ runs in 64 cells of 20. Of the 1,280 runs, 1,278 terminated on the model’s own stop decision and two on the cap of eight experiments. The study issued approximately 9,900 model calls and 4,768 estimator fits, the latter consuming 74 minutes of executor time in total.

3.3 Metrics

Four dependent variables are used.

Outcome metrics. The *gain over default* is the difference between the score of the model finally recommended by the autoresearch run and $a_0(d)$. It answers the primary question whether the tuning achieved anything that untuned defaults would not have given. A negative value indicates that the run terminated on a configuration inferior to untuned defaults.

The *improvement rate* is the proportion of executed trials whose score is strictly higher than the best score observed earlier in the same run.

Efficiency metrics. The *wasted trial ratio* is the number of proposals rejected before execution, together with those duplicating an earlier proposal in the same run, divided by the proposals attempted.

The *cost to best* is the number of executor calls made up to and including the call that produced the highest score of the run, taking the earliest such call on ties.

Measurement window. The stopping rule determines how many experiments a run performs, and every metric above aggregates over those experiments. Every metric is therefore computed over the first three experiments, this being the number performed at the fixed-budget level and hence the number available in every run.

3.4 Regression Model and Inference

Each component is coded as an indicator variable

$$X_i = \begin{cases} +1 & \text{if component } i \text{ is at level 1,} \\ -1 & \text{if at level 0,} \end{cases} \quad i = 1, \dots, 5, \quad (1)$$

and the outcome of one run is modeled as

$$Y = \beta_0 + \sum_{i=1}^5 \beta_i X_i + \sum_{j=2}^D \gamma_j D_j + \varepsilon, \quad (2)$$

where D_j indicates that the run was made on dataset j , the first dataset serving as the reference level. The γ_j are dataset fixed effects. They absorb the inherent difficulty of each task, its headroom above $a_0(d)$ and its response to the whitelist. Here β_0 denotes the mean across configurations on the reference dataset, β_i the effect of component i , and ε run-to-run error. $2\beta_i$ is the average change in Y produced by switching component i ; it is $2\hat{\beta}_i$ that is reported throughout. Because all thirty-two configurations are executed, the interaction model is also identified,

$$Y = \beta_0 + \sum_{i=1}^5 \beta_i X_i + \sum_{1 \leq i < j \leq 5} \beta_{ij} X_i X_j + \sum_{j=2}^D \gamma_j D_j + \varepsilon, \quad (3)$$

and this regression tests RQ2.

Estimation proceeds by ordinary least squares on the individual runs, with heteroskedasticity-consistent (HC3) standard errors and Benjamini–Hochberg adjustment within each metric.

3.5 Results

What the runs did. Before the estimates are presented, Table 2 describes the 1,280 runs themselves. In total the loop attempted 5,140 proposals, of which 4,768 were executed and 372 were rejected, and only five proposals in the whole study exactly duplicated an earlier one. Not one of the 5,140 replies was malformed JSON or of the wrong shape: 369 of the 372 rejections were whitelist violations, 365 of them on the random forest and four on gradient boosting, and the remaining three were estimator fits that failed. The hyperparameters the loop reached for were legitimate scikit-learn arguments that the whitelist does not admit, above all `min_samples_split` on the random forest, which accounts for 290 of the 369 violations, followed by `random_state` with 71. In this study the wasted trial ratio is therefore, almost entirely, the rate at which the loop names a hyperparameter outside an interface it has not been shown.

The best score arrived on the first executor call in 720 runs (56.2%), on the second in 291 (22.7%), on the third in 222 (17.3%), and on a later call in only 47 runs (3.7%); no run needed more than seven calls. Taken together, 96.3% of runs never improved after their third trial. On average 2.02 experiments per run were executed after the best had already been found, which amounts to 2,585 of the 4,768 executed trials, more than half of the compute in the study. On the gain over default, 40.0% of runs improved on a_0 , 39.1% recommended a configuration scoring exactly a_0 , and 20.9% terminated below it.

Main effects. Table 3 gives the main-effect estimates and Figure 2 plots them with their confidence intervals.

The wasted trial ratio is the most responsive metric. Four of the five components produce effects that survive adjustment, and the model accounts for $R^2 = 0.16$ of its variance. The reporting style produces the largest effect measured anywhere in the design, $2\hat{\beta}_O = -0.075$, corresponding to a reduction in the mean ratio from 0.106 at level 0 to 0.031 at level 1. Given the composition of the rejections described

Table 2: Upper panel: per-run quantities across all 1,280 runs. Lower panel: per-run means by level of the stopping rule S and the reporting style O . Study totals: 5,140 proposals attempted, 4,768 experiments executed, 372 proposals rejected, 5 duplicates.

Per run	Mean	Median	Max
Proposals attempted	4.02	3	9
Experiments executed	3.73	3	8
Proposals rejected	0.29	0	2
Duplicate proposals	0.004	0	1
Executor calls to best	1.71	1	7
<i>(complete runs; within the three-experiment window the maximum is 3)</i>			
Per run (mean)	$S = 0$ fixed	$S = 1$ adaptive	
Experiments executed	2.88	4.57	
Proposals rejected	0.17	0.41	
Experiments after best	1.30	2.73	
Per run (mean)	$O = 0$ terse	$O = 1$ full sentences	
Experiments executed	3.76	3.69	
Proposals rejected	0.43	0.15	

Table 3: Main-effect estimates $2\hat{\beta}_i$ per metric from equation (2), with heteroskedasticity-consistent standard errors and Benjamini–Hochberg adjusted p -values. The $\hat{\gamma}$ column is the fitted dataset fixed effect (Wine relative to Breast Cancer). Stars mark adjusted $p < 0.05$ (*), < 0.01 (**) and < 0.001 (***).

Metric	M	B	S	O	E	$\hat{\gamma}$	R^2
Gain over default	+0.001	-0.001	+0.001**	+0.000	+0.000	-0.001***	0.029
Wasted trial ratio	-0.023***	-0.034***	+0.044***	-0.075***	+0.005	-0.003	0.159
Improvement rate	+0.008	-0.026*	+0.034**	+0.006	+0.018	-0.064***	0.044
Cost to best	+0.06	-0.02	+0.05	-0.01	-0.11	-0.20***	0.025

above, this is a reduction in whitelist violations: 273 of the 369 occurred in runs with terse reporting and 96 in runs with full sentences. Search breadth and the evaluation criterion act in the same direction, with effects of -0.034 and -0.023 . The adaptive stopping rule acts in the opposite direction, $+0.044$, raising the mean ratio from 0.046 to 0.090. The mechanism is visible in the descriptives. The adaptive level raises the mean number of executed experiments from 2.88 to 4.57 and the mean number of rejected proposals per run from 0.17 to 0.41; the additional proposals do not duplicate earlier ones, of which there are only five in the study, but reach more often for a hyperparameter outside the whitelist, 260 of the 369 violations occurring under the adaptive rule.

The two outcome metrics, by contrast, respond to the stopping rule alone. It is the only component whose effect on the gain over default survives adjustment ($+0.001$, adj. $p = 0.005$), and it raises the improvement rate from 0.135 to 0.170. Search breadth is the only other component affecting the improvement rate, at -0.026 . No component affects the cost to best at the adjusted 0.05 level; the largest estimate is exploit-first at -0.11 , with adj. $p = 0.057$. The components that determine efficiency are therefore not the components that determine outcome.

On two of the four metrics the dataset fixed effect is roughly twice the largest component effect. Wine reduces the improvement rate by 0.064 and the cost to best by 0.20 relative to Breast Cancer, both at $p < 10^{-5}$, against largest component effects of 0.034 and 0.11 on the same metrics. Which task the loop is run on matters more for these two metrics than how the instruction file is worded, and this is exactly the variation that equation (2) removes from the residual.

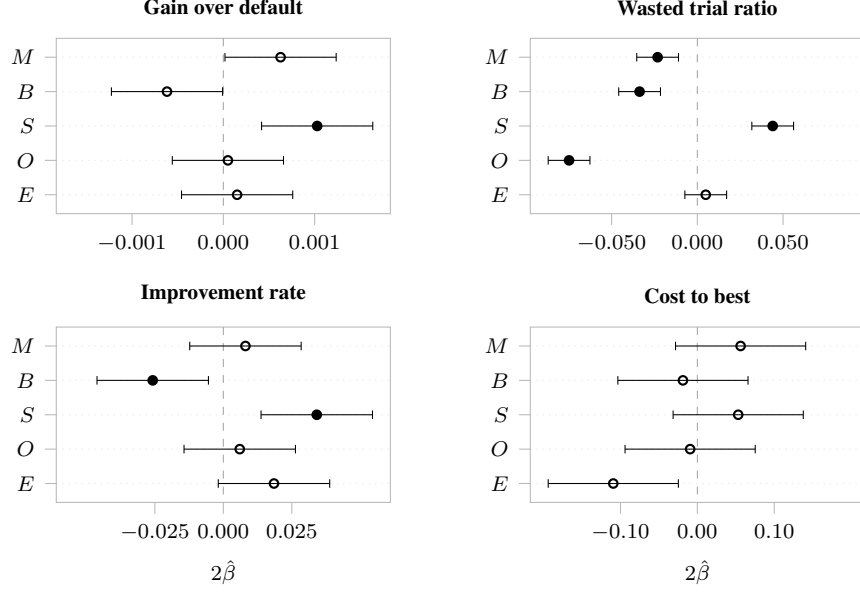


Figure 2: Estimated component effects on every metric. Points are $2\hat{\beta}_i$ from equation (2), bars are 95% intervals built from heteroskedasticity-consistent standard errors, and a filled point marks an effect that survives Benjamini–Hochberg adjustment at 0.05. The dataset fixed effect is in the model, so these are within-task effects. The four panels carry independent horizontal scales, the metrics being in different units.

Interactions. Table 4 gives the two-way interactions. Three of the forty estimated across the four metrics survive adjustment; for the remaining thirty-seven the data provide no evidence of interaction. All three occur on the wasted trial ratio and all three involve the reporting style: $S \times O$ at -0.031 , $B \times O$ at $+0.018$, and $O \times E$ at -0.016 . Of these, only $S \times O$ is comparable in magnitude to a main effect, and its sign indicates complementarity rather than substitution. The cell means make the direction plain: the adaptive budget raises the mean wasted trial ratio from 0.068 to 0.143 when reporting is terse, but only from 0.024 to 0.037 when it is in full sentences, so that the additive prediction for the adaptive-and-verbose cell, 0.100, overstates the observed 0.037 by a factor of nearly three. Equivalently, the reduction the reporting style buys is larger under an adaptive budget (-0.106) than under a fixed one (-0.043). A generous experiment budget is expensive in wasted proposals only when the loop is also reporting tersely. The remaining two interactions are smaller and of opposite character: $B \times O$ is the substitution case, both components reducing waste and each reducing it less when the other is already at level 1, while $O \times E$ resembles $S \times O$. Adding all ten interaction terms to the model of the wasted trial ratio raises R^2 only from 0.159 to 0.192, and the same three terms survive adjustment, with unchanged signs, when the ratio is refitted on the logit, square-root or arcsine-square-root scale, so the interaction structure is not an artifact of fitting a proportion on its natural scale. The components may therefore be treated as additive, which is the answer to RQ2, and the complete factorial is what permits the question to be tested rather than assumed.

Compliance. Table 4 also reports the share of runs whose behavior matched the instruction given, so that a component without effect can be told apart from one the model disregarded. B is checked by the number of model families a run proposes, S by the number of experiments it performs and E by whether its second proposal refines its first; O constrains wording rather than action, so it is given the mean characters of model-written text per step instead; M leaves no behavioral trace, because the executor computes the score whatever the instruction says. The fixed budget is followed in 89% of runs and the adaptive rule in only 20%, so the stopping-rule effects measure the effect of offering an adaptive budget rather than of a loop that reliably stops on diminishing returns, and every estimate in Table 3 is

Table 4: Two-way interaction estimates $2\hat{\beta}_{ij}$ from equation (3), and compliance per component. The four interactions with the smallest adjusted p are shown, with adjusted p given directly; compliance is reported as level 0 / level 1.

Interaction	Estimate	adj. p	Component	Compliance
$S \times O$ (wasted trial ratio)	-0.031	0.000	B	0.66 / 0.82
$B \times O$ (wasted trial ratio)	+0.018	0.009	S	0.89 / 0.20
$O \times E$ (wasted trial ratio)	-0.016	0.019	E	0.88 / 0.69
$M \times B$ (wasted trial ratio)	-0.010	0.171	O	205 / 386 chars
			M	not observable

an intention-to-treat estimate.

4 Conclusion

In an autoresearch loop, `program.md` performs the function that controller code performs in conventional software, yet it cannot be read in the manner of code. This thesis has treated it as a configuration vector. Five components of `program.md` were written at two levels each, all $2^5 = 32$ configurations were generated and executed against a real executor on two datasets with twenty replicates apiece, and the resulting behavior of the 1,280 runs was reduced to four metrics and regressed on an orthogonal ± 1 coding of the components. Because the configurations were constructed rather than collected, the coefficients admit a causal reading. Because the factorial is complete, the independence of the components could be tested rather than assumed.

4.1 Contributions

The main contributions of this thesis are as follows.

1. **A method for attributing the behavior of an autoresearch loop to its instruction file.** The instruction file is partitioned into slots, each slot is written at two levels by a model distinct from the agent, the complete factorial is executed, and the effects are estimated by a regression whose orthogonal coding and dataset fixed effect follow from the design. The method is not tied to the five components varied here.
2. **An answer to RQ1.** The components determine the process, not the outcome. The reporting style produced the largest effect in the design, reducing the mean wasted trial ratio from 0.106 to 0.031, and search breadth and the evaluation criterion moved the same metric by -0.034 and -0.023 . Almost every wasted trial was a proposal naming a hyperparameter outside an undisclosed whitelist, and none was a malformed reply. The stopping rule was the only component with an effect on either outcome metric, $+0.001$ on the gain over default and $+0.034$ on the improvement rate, and no component determined the cost to best. On two of the four metrics the dataset fixed effect exceeded every component effect, which is the reason equation (2) controls for it.
3. **An answer to RQ2.** Thirty-seven of the forty two-way terms show no evidence of interaction. The three that survive adjustment are all on the wasted trial ratio and all involve the reporting style, and together the ten interaction terms raise R^2 on that metric only from 0.159 to 0.192, so the components may be treated as additive.

With untuned defaults already at $a_0 \approx 0.96$, the instruction file determines how the loop conducts its search rather than what the search returns.

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A The files the loop runs on

Section 1 describes the three-file contract of Karpathy’s autoresearch: `prepare.py` fixes the data and the score, `train.py` holds the model and the training loop, and `program.md` holds the instructions. The study keeps that contract file for file. All three are reproduced below: the specification of `program.md` in Appendix B, `prepare.py` in Appendix C, and `train.py` together with the executor whitelist in Appendix D.

One thing does differ, and it is the narrowing of Section 2 rather than a departure from the contract. In Karpathy’s loop the agent edits `train.py` directly. Here the search space is a hyperparameter space, not arbitrary code, so that the instruction file stays the only free variable; the agent names a model family and hyperparameter values in a JSON proposal and `train.py` builds exactly that estimator. The agent therefore acts on `train.py` through a fixed interface instead of rewriting it. What is unchanged is the ownership that matters for measurement: `prepare.py` is reachable by nobody, so no level of any component can alter what a score means, and every number in a transcript is a measurement rather than a claim.

B Specification of the instruction file

A `program.md` is not written; it is rendered. This appendix gives the whole specification, from which any of the thirty-two files can be reconstructed exactly: the template, which is fixed, and the ten paragraphs that fill its five slots.

Appendix B.1 gives the template. Everything in it is byte-identical across all thirty-two configurations of a task. The five marked slots are the independent variables. The three remaining substitutions, `task_title`, `research_question` and `dataset_name`, are set by the task and not by the configuration, so they are constant across the thirty-two files of one dataset and differ only between datasets.

Appendix B.2 gives the ten paragraphs, two per slot, verbatim. A configuration $c = (c_1, \dots, c_5)$ selects one paragraph per slot, and substituting one paragraph for the other level of the same slot is the only difference between two files. These ten texts are the treatment: they were written once by a language model from fixed meta-prompts that were not preserved, then frozen and reused word for word, for the reason given in Section 2. Table 1 summarizes them; what follows is the text itself, since it is the wording rather than the summary that the loop responds to.

B.1 Template

```
# Research Task: {{ task_title }}
```

You are an autonomous research agent. Your task is to investigate the following question through real, executed experiments (not simulated ones):

```
> {{ research_question }}
```

You have access to a fixed dataset ({{ dataset_name }}, already loaded and split into train/validation) and to a tool that fits either a RandomForestClassifier or a GradientBoostingClassifier with hyperparameters you choose. The tool always returns the same record: 5-fold cross-validated accuracy on the training split, its standard deviation across folds, and accuracy on the held-out validation split.

```
## Evaluation

< slot M --- metric >
{{ metric_text }}
```

```
## Search strategy

< slot B --- breadth >
{{ breadth_text }}
```

```
< slot E --- emphasis >
{{ emphasis_text }}
```

```
## Budget and stopping

< slot S --- stopping >
{{ stopping_text }}
```

```
## Reporting style

< slot O --- output format >
{{ output_format_text }}
```

```
## Required response format
```

At every step, respond with a single JSON object and nothing else, matching one of these two shapes:

Proposing an experiment:

```
```json
{"action": "propose", "hypothesis": "<why you think this configuration will help>",
 "model": "random_forest" | "gradient_boosting",
 "params": {"<hyperparameter>": <value>, ...}}
```
```

Interpreting the last result and deciding what happens next:

```
```json
{"action": "interpret", "interpretation": "<what the result tells you>",
 "decision": "continue" | "stop",
 "final_recommendation": "<only required when decision == stop: your final recommended model and hyperparameters, and why>"}
```
```

B.2 The ten treatment paragraphs

slot M --- Evaluation criterion (substituted for {{ metric.text }})
level 0: Evaluate model quality in whatever way you judge appropriate.

level 1: Evaluate every model using 5-fold cross-validated accuracy on the training split, AND report the standard deviation across folds as a measure of stability. A configuration is only better than another if its mean accuracy is higher; use the standard deviation only to break ties or to flag instability.

slot B --- Search breadth (substituted for {{ breadth.text }})
level 0: Stay within a single model family per experiment. Pick either RandomForestClassifier or GradientBoostingClassifier for the whole session unless you have exhausted reasonable hyperparameters for it and explicitly decide to switch.

level 1: You are free to compare RandomForestClassifier and GradientBoostingClassifier side by side within the same session, and to vary multiple hyperparameters of each at once.

slot S --- Stopping rule (substituted for {{ stopping.text }})
level 0: You have a fixed budget of exactly 3 experiments in total. Run exactly 3, then stop and report your final recommendation, regardless of the outcome of experiment 3.

level 1: You have a budget of up to 8 experiments. After each experiment, decide whether to continue: stop as soon as an experiment fails to improve the evaluation metric (defined above) by more than 0.002 over the best result so far, or when you reach 8 experiments, whichever comes first.

slot O --- Reporting style (substituted for {{ output.format.text }})
level 0: Keep every response short: state the action you are taking and the numbers involved, with minimal prose.

level 1: Explain your reasoning at every step in full sentences: what you hypothesize, why, what you expect to observe, and how you interpret the actual result relative to that expectation.

slot E --- Search emphasis (substituted for {{ emphasis.text }})
level 0: Prioritize breadth of search: try several meaningfully different configurations before you spend more than one experiment refining any single one of them.

level 1: Prioritize depth: as soon as one configuration looks promising, immediately refine it with nearby variations (e.g. slightly different hyperparameter values) before trying anything unrelated.

C prepare.py

The data and the score. Nobody edits this file, and the agent never sees it, so no level of any component can change what an experiment is measured on or how it is scored. One stratified 70/30 partition per task, fixed with the same seed for every run, and one scoring procedure returning the same record every time, whatever a configuration's evaluation paragraph emphasizes.

```
"""
prepare.py -- the data and the score. Nobody edits this file.

The first of the three files of the autoresearch contract (thesis, Introduction).
It fixes what every experiment is measured on and how it is scored, so that a
score is comparable across configurations, across replicates and across the whole
study. Neither the agent nor the instruction file can reach it: the agent never
sees this module, and no level of any component in program.md changes what it
returns. That is what makes a number in a transcript a measurement rather than a
claim.

Two responsibilities, and no others:

    load(dataset)           the fixed train/validation partition for one task
    score(estimator, dataset) the record every experiment is scored by

Every task uses the identical split protocol: one stratified 70/30 partition
fixed with the same seed across all runs, so the split is a constant of the study
rather than a source of variation (thesis, Experimental setup). The score is the
same record whatever a configuration's evaluation paragraph emphasises:
cross-validated accuracy on the training split, its across-fold standard
deviation, and accuracy on the held-out validation split.

Adding a dataset here does not put it in the study; the suite is whatever
study_config.DATASETS says.
"""
import sys
from functools import lru_cache
```

```

from pathlib import Path

from sklearn import datasets as skds
from sklearn.model_selection import cross_val_score, train_test_split

sys.path.insert(0, str(Path(__file__).resolve().parent / "config"))
from study_config import CV_FOLDS, SCORING, SPLIT_SEED, TEST_SIZE # noqa: E402

# name -> zero-argument loader returning (X, y)
LOADERS = {
    "breast_cancer": lambda: (lambda d: (d.data, d.target))(skds.load_breast_cancer()),
    "wine": lambda: (lambda d: (d.data, d.target))(skds.load_wine()),
    "digits": lambda: (lambda d: (d.data, d.target))(skds.load_digits()),
    "iris": lambda: (lambda d: (d.data, d.target))(skds.load_iris()),
}

@lru_cache(maxsize=None)
def load(name):
    """Return the fixed split for one dataset as (X_train, X_val, y_train, y_val)."""
    if name not in LOADERS:
        raise KeyError(f"unknown dataset {name!r}; known: {sorted(LOADERS)}")
    X, y = LOADERS[name]()
    return train_test_split(
        X, y, test_size=TEST_SIZE, random_state=SPLIT_SEED, stratify=y
    )

def score(estimator, dataset):
    """The one scoring procedure of the study. Returns the record, or raises."""
    X_tr, X_va, y_tr, y_va = load(dataset)
    cv = cross_val_score(estimator, X_tr, y_tr, cv=CV_FOLDS, scoring=SCORING)
    estimator.fit(X_tr, y_tr)
    val = estimator.score(X_va, y_va)
    return {
        "cv_accuracy_mean": round(float(cv.mean()), 6),
        "cv_accuracy_std": round(float(cv.std()), 6),
        "val_accuracy": round(float(val), 6),
    }

def describe(name):
    import numpy as np
    X_tr, X_va, y_tr, y_va = load(name)
    return {
        "dataset": name,
        "n_train": int(X_tr.shape[0]),
        "n_val": int(X_va.shape[0]),
        "n_features": int(X_tr.shape[1]),
        "n_classes": int(len(np.unique(y_tr))),
    }

```

D train.py and the executor whitelist

The model and the training loop. The whitelist here is the boundary of the search space: a proposal naming an unknown family, or a hyperparameter outside the whitelist for its family, is rejected before it reaches the scoring procedure, so a malformed proposal costs a turn but not an experiment. The whitelist is defined once in the study configuration and is the same on every task; it is not shown to the agent, whose instruction file only says that the hyperparameters are its to choose.

```

# config/study_config.py (excerpt): the executor whitelist
ALLOWED_PARAMS = {
    "random_forest": {"n_estimators", "max_depth", "min_samples_leaf", "max_features"},
    "gradient_boosting": {"n_estimators", "max_depth", "learning_rate", "subsample"},
}

"""
train.py -- the model and the training loop.

```

The second file of the autoresearch contract (thesis, Introduction). In Karpathy's loop this is the file the agent edits, and it is the only one it may edit. The loop studied here narrows the search space from arbitrary code to a hyperparameter space, so that the instruction file stays the only free variable (thesis, The loop under study). The narrowing changes how the agent reaches this file, not which file it acts on: instead of rewriting the source, the agent names a model family and hyperparameter values in a JSON proposal, and this module

builds and trains exactly that estimator. Everything the agent can vary about the model lives here; nothing it can vary about the measurement does, that being `prepare.py`.

The whitelist is the boundary of the search space. A proposal naming an unknown family, or a hyperparameter outside the whitelist for its family, is rejected here and never reaches the scoring procedure, which is what keeps a malformed proposal from costing an experiment.

```

    run_experiment(model, params, seed, dataset) -> record | {"error": ...}
"""
import sys
import time
from pathlib import Path

from sklearn.ensemble import GradientBoostingClassifier, RandomForestClassifier

STUDY = Path(__file__).resolve().parent
sys.path.insert(0, str(STUDY))
sys.path.insert(0, str(STUDY / "config"))
import prepare # noqa: E402
import study_config # noqa: E402 (read live, so WORKERS can set EXECUTOR_N_JOBS)
from study_config import ALLOWED_PARAMS # noqa: E402

def build(model, params, seed):
    """The estimator named by a proposal. Raises on an unknown family."""
    if model == "random_forest":
        # n_jobs is set to 1 when the outer loop is parallelised, so that the
        # forest's internal threads do not oversubscribe the cores.
        return RandomForestClassifier(
            random_state=seed, n_jobs=study_config.EXECUTOR_N_JOBS, **params)
    if model == "gradient_boosting":
        return GradientBoostingClassifier(random_state=seed, **params)
    raise ValueError(f"unknown model {model!r}")

def validate(model, params):
    """Whitelist check. Returns an error string, or None if the proposal is legal."""
    if model not in ALLOWED_PARAMS:
        return f"unknown model '{model}', must be one of {sorted(ALLOWED_PARAMS)}"
    bad = set(params) - ALLOWED_PARAMS[model]
    if bad:
        return f"unsupported params for {model}: {sorted(bad)}"
    return None

def run_experiment(model, params, seed, dataset):
    """Train one estimator on one dataset and score it. Record, or {'error': ...}."""
    err = validate(model, params)
    if err:
        return {"error": err}

    t0 = time.time()
    try:
        record = prepare.score(build(model, params, seed), dataset)
    except Exception as exc:
        # noqa: BLE001 - reported to the model
        return {"error": f"experiment failed: {exc}"}

    return {"model": model, "params": params, "dataset": dataset, **record,
            "wall_time_s": round(time.time() - t0, 3)}
```